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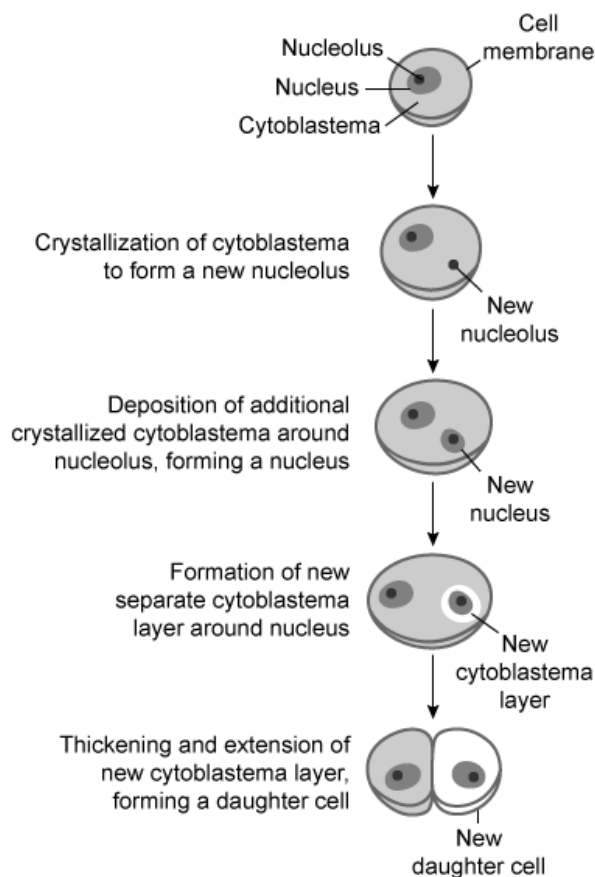


Figure 1 Early hypothesis of animal cell reproduction

Although this hypothesis would later be refuted, observations of animal cells during these early studies also indicated that individual cell components tended to remain within the boundary of the cell, providing evidence for the presence of an outer plasma membrane (PM). Some researchers initially thought that the PM was a main site of metabolic processes. Phospholipids were later identified as the major component of PMs, but the arrangement of phospholipids into a bilayer became clearer with additional experiments using mammalian erythrocytes.

In these erythrocyte experiments, all membrane lipids were extracted from a cell and arranged as a monolayer. The total surface area (SA) of this lipid monolayer was measured along with the total SA of the erythrocyte. Some studies found the ratio of lipid SA to total erythrocyte SA to be 2:1 whereas other studies calculated a

ratio of approximately 1:1.

Additional research found that the crossing of hydrophilic molecules through hydrophobic regions of lipid-based membranes was energetically unfavorable. Therefore, many experiments also focused on the transport of hydrophilic molecules (eg, ions) across PMs. In the 1950s, two mechanisms for ion transport across membranes were hypothesized (Figure 2).

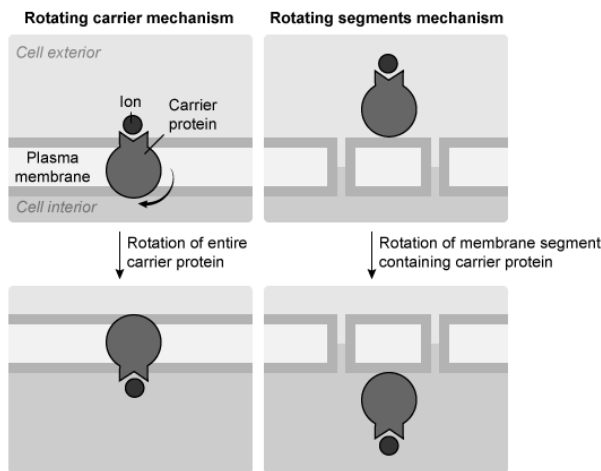


Figure 2 Hypothetical mechanisms of membrane ion transport

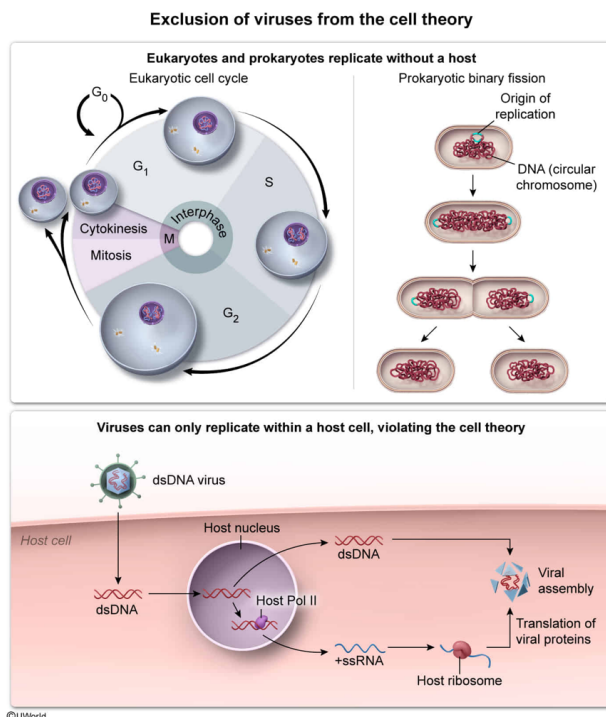
Viruses do not exhibit characteristics of cells as defined by the cell theory. The most likely reason why viruses are excluded from the cell theory is that:

- ☐ A. viruses have very small genomes compared to prokaryotes and eukaryotes. [1%]
- ☐ B. viruses parasitize host organisms to cause disease. [7%]
- ☒ C. viruses utilize host cell enzymes for reproductive processes. [83%]
- ☐ D. viruses copy their genomes using reverse transcription. [7%]

Correct

83% answered correctly

Explanation:



Initial ideas of the **cell theory** were formulated in the 1600s by Robert Hooke and Anton van Leeuwenhoek, who were the first scientists to observe cells using microscopes. Today, cell theory includes several basic principles:

- Cells comprise the structures of all living organisms.
- Within organisms, each cell serves as an individual unit of life.
- To reproduce, cells duplicate themselves via cell division (ie, **binary fission** in **prokaryotes** and the **cell cycle** in **eukaryotes**).
- All cells have genetic information that is encoded in **DNA**, which can be passed from parent cells to daughter cells.

Viruses differ from living cells because they are **obligate intracellular parasites** that cannot reproduce outside of a host cell. This dependence is because viruses *lack* the **transcriptional** and **translational** machinery (eg, RNA polymerase, ribosomes) present in all prokaryotic and eukaryotic cells, and therefore must utilize host cell machinery to replicate. In addition, although the **genomes** of all prokaryotes and eukaryotes consist of DNA, viruses may contain either **DNA** or **RNA** genomes.

The cell theory states that cells reproduce by duplicating *themselves* via division (ie, 微信Mai_ange

without the need of a host cell). Therefore, viruses are **excluded** from the **cell theory** because they must **utilize host cell enzymes** for **reproductive processes** (ie, they *cannot* duplicate themselves outside of a host).

(Choice A) Genome size *varies* among cells and viruses. Although viruses tend to have very small genomes, genome size is *not* a principle described by the cell theory.

(Choice B) Although viruses can parasitize host organisms and cause disease, disease-causing eukaryotic and prokaryotic parasites (eg, some protozoans, bacteria) also exist. Therefore, this trait is *not* a unique characteristic of viruses that causes them to be excluded from the cell theory.

(Choice D) Certain viruses (ie, **retroviruses**) copy their genomes using a specialized enzyme called reverse transcriptase. However, reverse transcription can also occur in prokaryotic and eukaryotic genomes (ie, in **retrotransposons**), and reverse transcription is *not* a principle included in the cell theory.

Educational objective:

The cell theory consists of several key principles: Cells comprise the structures of all living organisms and are individual units of life; cells reproduce by dividing themselves (ie, via binary fission in prokaryotes and the cell cycle in eukaryotes); and cell genomes are composed of DNA that can be passed from parent cells to daughter cells.

Time spent:0 sec

Subject:
Biology

QID:403340

System:
Cellular Biology

In the 1800s, cell theory studies included research into cell reproduction and major cell structures. Specifically, an early hypothesis regarding animal cell reproduction predicted that all cellular components arose from a fluid substance called cytotlastema. It was thought that during reproduction, a portion of an existing cell's cytotlastema would crystallize into a dense nucleolus, followed by the deposition of additional crystallized cytotlastema to form a new nucleus. Researchers believed that a new, separate cytotlastema layer would then thicken around this new nucleus, ultimately forming a daughter cell (Figure 1).

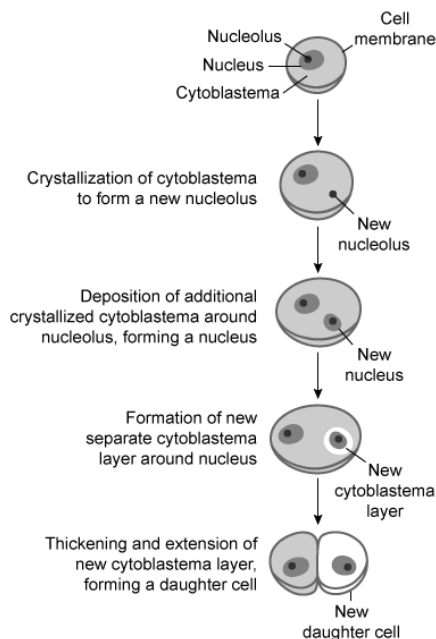


Figure 1 Early hypothesis of animal cell reproduction

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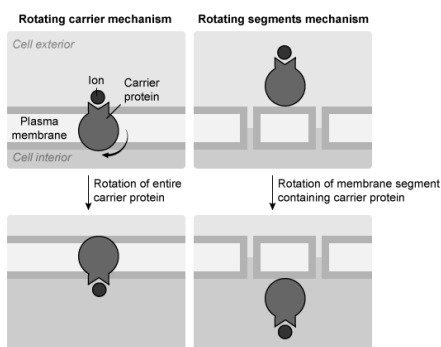


Figure 2 Hypothetical mechanisms of membrane ion transport

Which of the following observations of the cell cycle is similar to a characteristic of the animal cell reproduction hypothesis shown in Figure 1?

- I. Nuclear materials are replicated from existing nuclear materials
- II. A cleavage furrow forms as cytoplasmic division occurs in a parent cell
- III. Two individual nuclei form within a single parent cell during mitosis

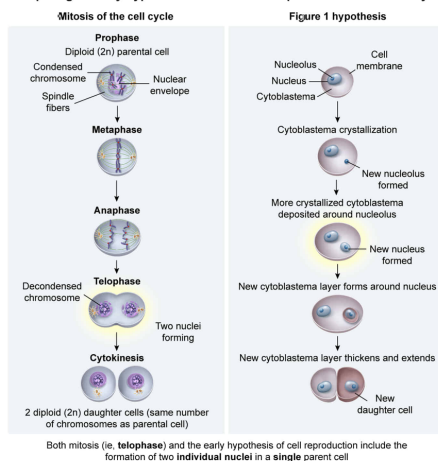
- ☐ A. I only [5%]
- ✓ ☒ B. III only [10%]
- ☐ C. I and II only [39%]
- ☐ D. II and III only [44%]

Correct

10% answered correctly

Explanation:

Comparing an early hypothesis of animal cell reproduction to the cell cycle



One focus of the **cell theory** is cellular reproduction, which is now understood to occur via the **cell cycle** in **eukaryotic cells**. Following cell growth and **DNA replication** during interphase of the cell cycle, eukaryotic cells divide via **mitosis**, which includes the following phases:

1. **Prophase**: DNA condenses to form chromatids. Each pair of **sister (identical) chromatids** are joined by a centromere to form chromosomes. The nuclear envelope breaks down and centrosomes (microtubule-organizing structures) move to opposite poles within the cell. Microtubules grow from centrosomes, forming the mitotic spindle.
2. **Metaphase**: Spindle fiber microtubules attach to chromosomes at **kinetochores**, and chromosomes align at the metaphase plate.
3. **Anaphase**: Sister chromatids are pulled apart by spindle fibers, migrating to opposite cellular poles to form two separate sets of chromosomes within the cell.
4. **Telophase**: A nuclear envelope forms around each set of chromosomes in the cell. The chromosomes decondense.

Following telophase, cytokinesis (cytoplasmic division) of the parent cell forms two identical daughter cells.

Figure 1 shows an early animal cell reproduction hypothesis in which cells were thought to arise from a substance called cytoplasm. The hypothesis was new cells formed when existing cytoplasm crystallized (ie, hardened), forming structures (eg, **nucleolus**, nucleus) necessary to generate a new daughter cell.

The question involves observations of the cell cycle that are *similar* to a characteristic of this early hypothesis. In Figure 1, *two* individual nuclei were thought to form within one parental cell via cytoplasm crystallization. This characteristic is most **similar** to mitotic **telophase** when nuclear envelopes reform around two separate sets of chromosomes, forming **two** individual nuclei in a **single** parental cell (**Number III**).

(Number I) During the cell cycle, nuclear materials (eg, DNA) are replicated from existing nuclear materials (eg, parental DNA strands). However, in Figure 1 the new nucleus forms from cytotblastema (*not* the existing nucleus).

(Number II) During telophase and cytokinesis, a **cleavage furrow** forms as a contractile actin filament ring begins pinching a parent cell in two, ultimately dividing the cytoplasm into two daughter cells. However, in Figure 1 a daughter cell arises when a new cytotblastema layer thickens around a newly formed nucleus (*not* via cleavage furrow formation).

Educational objective:

During mitotic prophase, the parent cell nuclear envelope breaks down and chromosomes condense. Chromosomes then align and sister chromatids separate during metaphase and anaphase, respectively. Lastly, nuclear envelopes reform around each chromosome set during telophase, and cytokinesis (cytoplasmic division) forms two identical daughter cells.

Time spent:0 sec

Subject:
Biology

QID:403341

System:
Cellular Biology

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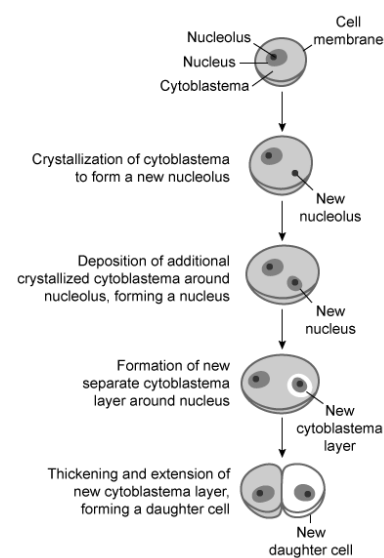


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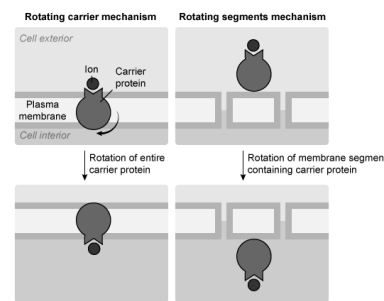
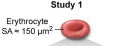
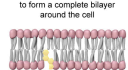
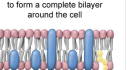


Figure 2 Hypothetical mechanisms of membrane ion transport

Based on the passage experiments testing membrane lipid surface area (SA) in erythrocytes, which of the following could best explain a 1:1 ratio of membrane lipid SA to total erythrocyte SA?

- ☐ A. Erythrocytes are generally smaller in size than other mammalian cell types. [11%]
- ☐ B. Erythrocytes lack internal membrane-bound organelles. [53%]
- ☒ C. Erythrocytes may have nonlipid structures in their plasma membranes. [25%]
- ☐ D. Erythrocytes also have cholesterol in their cell membranes. [9%]

Explanation:

Study of erythrocyte membrane structure			
	Study 1	Study 2	
1. Extraction of membrane lipids			
2. Arrangement of membrane lipids into a monolayer			
3. Calculate SA ratio = SA of membrane lipids / Total SA of erythrocyte	$\frac{300 \mu\text{m}^2}{150 \mu\text{m}^2} = \frac{2}{1}$	$\frac{150 \mu\text{m}^2}{150 \mu\text{m}^2} = \frac{1}{1}$	
4. Evaluate results	 2:1 ratio enough phospholipids present to form a complete bilayer around the cell	 1:1 ratio not enough phospholipids present to form a complete bilayer around the cell Membrane proteins (ie, nonlipid components) fill space not occupied by phospholipids	

Cell plasma membranes (PMs), described by the **fluid mosaic model**, are composed mainly of a **phospholipid bilayer** along with various proteins and other components (eg, **cholesterol**) interspersed throughout this bilayer. Within the phospholipid bilayer, **hydrophilic** (ie, "water-attracting") phospholipid heads face the extracellular and intracellular aqueous environments, and **hydrophobic** (ie, "water-repelling") tails are found in the interior, nonaqueous portion of the membrane.

The passage describes experiments in which all membrane lipids were extracted from an **erythrocyte** (red blood cell) and arranged as a monolayer. The surface area (SA) of the lipid monolayer was measured along with the total SA of the erythrocyte. Some studies found that the ratio of lipid SA to total erythrocyte SA was 2:1, indicating that the lipid SA necessary to form the membrane surrounding the cell is double the SA of the cell itself (ie, the lipids are arranged in a **bilayer**).

However, other studies calculated the ratio of lipid SA to total erythrocyte area as **1:1**. Because membrane phospholipids are now known to be arranged in a bilayer, a ratio of 1:1 would indicate that **fewer** phospholipids are present than would be required to form a complete bilayer around the cell. Therefore, erythrocytes must have other **nonlipid structures** in their membranes (ie, proteins) that also contribute to the SA of the PM surrounding the cell.

(Choices A and B) Regardless of their size or the fact that they do not possess internal **membrane-bound organelles**, the phospholipids in erythrocyte PMs are arranged as a bilayer. In the absence of other PM components (eg, proteins), the SA of membrane lipids arranged as a monolayer should be approximately double the total SA of the erythrocyte (ie, a 2:1 ratio). Therefore, cell size or lack of **internal** membranes alone does **not** explain a 1:1 ratio of membrane lipid SA to total erythrocyte SA.

(Choice D) Cholesterol is another type of lipid found in cell membranes. The presence of cholesterol should **increase** the SA of lipids present in the erythrocyte membrane and would **not** explain a **lower** ratio of lipid SA to total erythrocyte SA.

Educational objective:

Plasma membranes of cells are described as a fluid mosaic composed mainly of a phospholipid bilayer with other components (eg, membrane proteins, cholesterol) interspersed throughout.

Time spent: 0 sec
Subject:
Biology

QID: 403342
System:
Cellular Biology

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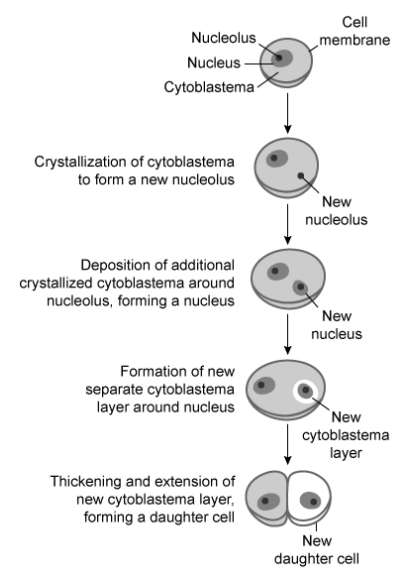


Figure 1 Early hypothesis of animal cell reproduction

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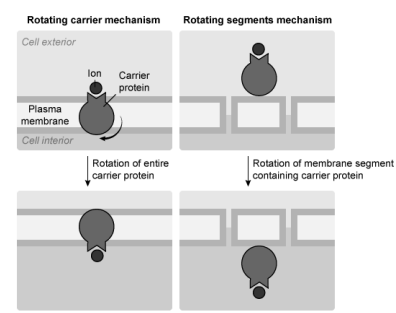


Figure 2 Hypothetical mechanisms of membrane ion transport

Which of the following experimental approaches would be most effective to determine if a plasma membrane protein operates as a rotating carrier during ion transport, as shown in Figure 2?

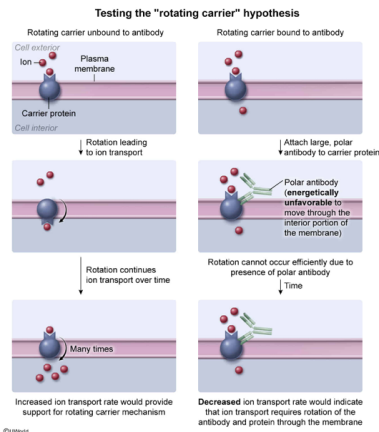
- ☐ A. Identifying the types of amino acids present in outer and inner regions of the carrier protein to determine their ability to bind ions for transport. [9%]
- ☐ B. Mutating the gene encoding this carrier protein and testing if ion transport still occurs. [15%]
- ☐ C. Determining if phospholipids near the carrier protein on the exterior side of the membrane move to the interior side of the membrane during ion transport. [43%]

- ✔ ☒ D. Attaching a large polar antibody to the portion of the carrier protein facing the cell exterior and observing the effect on ion transport rate. [32%]

Correct

31% answered correctly

Explanation:



According to the passage, research confirmed that the crossing of **hydrophilic** ("water-attracting") molecules through the hydrophobic ("water-repelling") regions of **lipid-based cell membranes** is **energetically unfavorable**.

This movement is unfavorable because hydrophilic molecules (eg, **polar molecules**, ions) normally interact with **water molecules** via **hydrogen bonding** or **ion-dipole interactions**. These interactions must be broken for hydrophilic molecules to enter the hydrophobic portion of the cell membrane (ie, the region that *cannot* strongly interact with hydrophilic molecules). Breaking these hydrophilic interactions requires energy and is therefore *unfavorable*. **Transport proteins**, which can facilitate either **active** or **passive** transport, allow hydrophilic molecules to cross the membrane by promoting **energetically favorable** interactions that release energy.

Figure 2 shows two hypothetical ion transport mechanisms. The "rotating carrier" mechanism hypothesizes that a transmembrane protein binds an extracellular ion and then rotates to deliver this ion intracellularly. For a rotating carrier, all protein components (including the exterior ion-binding portion) must rotate through the hydrophobic portion of the cell membrane. To determine if a membrane protein operates via this mechanism, a **large polar antibody** could be **attached** to the portion of the protein facing the cell exterior. Because polar molecules (ie, antibodies) traversing the hydrophobic cell membrane region are energetically *unfavorable*, the ion transport rate should *decrease* if the protein is a rotating carrier.

The following experiments would *not* provide evidence that a membrane protein is a rotating carrier:

- Determining the ability of the protein's **amino acids** to bind ions for transport would only provide information about the protein's functional binding, *not* how these bound ions *move* through the membrane via the protein (**Choice A**).
- Mutating** the gene encoding the protein would only provide information about how the mutation affects protein function, *not* how the wild-type protein moves across a membrane (**Choice B**).
- The "rotating segments" mechanism proposed in Figure 2 hypothesizes that an entire membrane segment (ie, phospholipids) rotates to move a carrier protein and bound ion intracellularly. Therefore, phospholipids near the protein on the exterior side of the membrane moving to the interior side during ion transport would provide evidence of a "rotating membrane segment," *not* a "rotating carrier protein" (**Choice C**).

Educational objective:

Because interactions between hydrophobic and hydrophilic molecules are energetically unfavorable, hydrophilic molecules (eg, polar molecules, ions) require transport proteins to cross the interior hydrophobic (nonpolar) portion of the plasma membrane.

Time spent: 0 sec
Subject:
Biology

QID: 403343
System:
Cellular Biology

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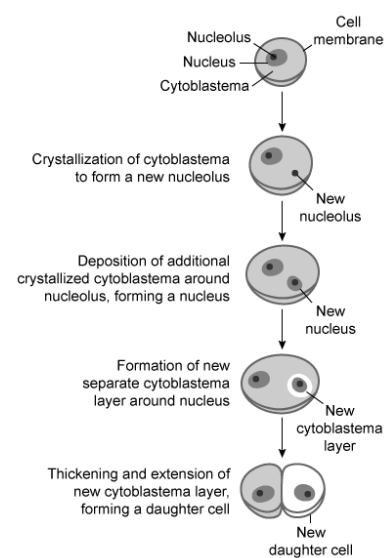


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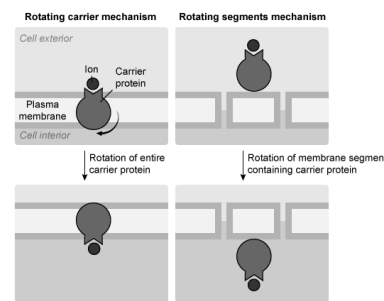


Figure 2 Hypothetical mechanisms of membrane ion transport

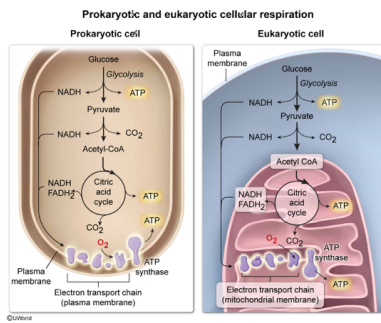
Given the early hypothesis that the plasma membrane (PM) serves as a site of metabolism in cells, are there any types of cells in which this hypothesis can be considered true based on current scientific knowledge?

- ☐ A. Yes; archaea are unique from other domains of life in that they can perform anaerobic metabolic processes using molecules located in the PM. [15%]
- ☐ B. No; the breakdown of glucose via glycolytic enzymes occurs in the cytoplasm of all known cell types. [11%]
- ☒ C. Yes; prokaryotic cells express electron transport chain proteins in their PMs despite lacking membrane-bound organelles. [67%]
- ☐ D. No; aerobic cellular respiration in the mitochondria produces the greatest amount of ATP. [5%]

Correct

67% answered correctly

Explanation:



Both **eukaryotic** and **prokaryotic** organisms catabolize glucose to produce ATP via **cellular respiration** or **fermentation**. Both **metabolic** processes start with **glycolysis**, a process occurring in the cytosol in which a single molecule of glucose is broken down into two pyruvate molecules.

In eukaryotic cellular respiration, pyruvate is imported into mitochondria where it is converted into acetyl-CoA and enters the Krebs (citric acid) cycle. The Krebs cycle generates the **electron carriers** NADH and FADH₂. Electrons from NADH and FADH₂ are then

passed to the **electron transport chain** (ETC). As electrons are moved along the ETC, ETC protein complexes pump protons across the inner mitochondrial membrane to generate a **proton gradient**. Protons then flow down this gradient through the enzyme **ATP synthase**.

In contrast, prokaryotic cells (ie, **both bacteria and archaea**), which do *not* possess **membrane-bound organelles** (eg, mitochondria), perform the Krebs cycle in the cytoplasm and have ETCs located on their outer PMs (**Choice A**).

In the passage, an early researcher hypothesized that the PM serves as a site of metabolism in cells. Because prokaryotes express **ETC** proteins in their **PMs**, this hypothesis can be considered **true** in the context of prokaryotic cells (ie, metabolic reactions of the ETC occur on the PMs of prokaryotes).

(Choice B) Although glycolysis is a **cytosolic** process in **all organisms** (ie, prokaryotes and eukaryotes), ETC proteins in the PMs of prokaryotes provide **support** for the hypothesis that the PM is a site of metabolism.

(Choice D) Aerobic respiration via the Krebs cycle and ETC produces *more* ATP than anaerobic processes such as fermentation. These aerobic processes occur in the mitochondria of eukaryotic cells. However, prokaryotic cells have ETCs in their PMs, indicating that the PM is a site of metabolism in these organisms.

Educational objective:

Both eukaryotic and prokaryotic cells carry out glycolysis in the cytoplasm. In eukaryotes, pyruvate molecules resulting from glycolysis are transported into the mitochondria, membrane-bound organelles where reactions of the Krebs cycle and electron transport chain (ETC) occur. In contrast, prokaryotic cells carry out the Krebs cycle in their cytoplasm and have ETC proteins embedded in the outer plasma membrane.

Time spent: 0 sec

Subject:
Biology

QID: 403344

System:
Cellular Biology

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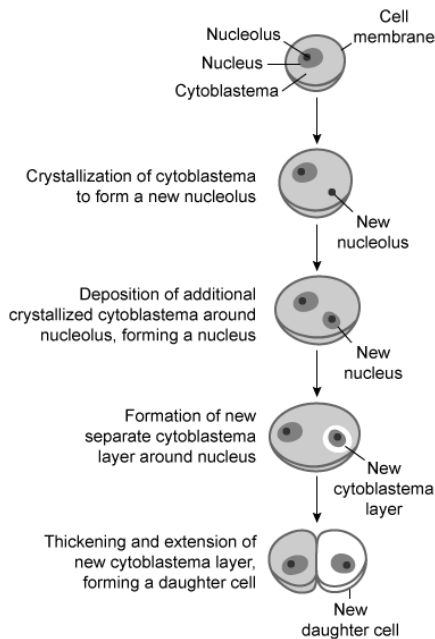


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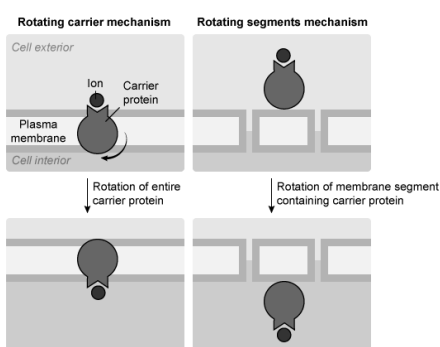
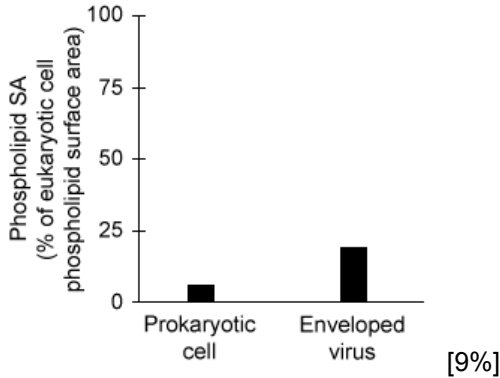


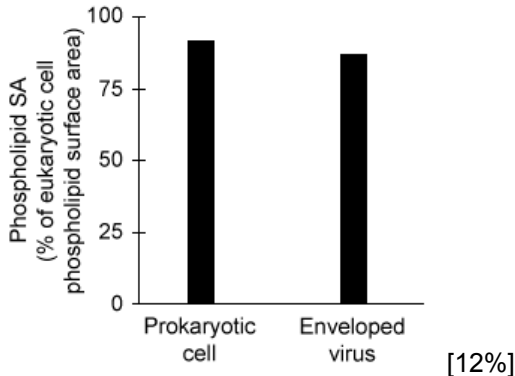
Figure 2 Hypothetical mechanisms of membrane ion transport

Phospholipid surface areas (SAs) were determined for multiple cell types and enveloped viruses using the methods described in the passage. Given this data, which of the following graphs illustrates the most likely phospholipid SAs of a prokaryotic cell and an enveloped virus relative to the phospholipid SA of a eukaryotic cell?

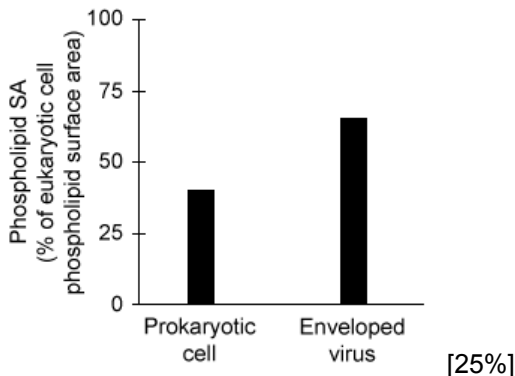
☐ A.



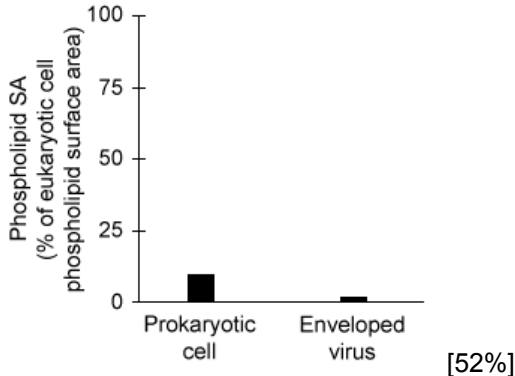
☐ B.



☐ C.

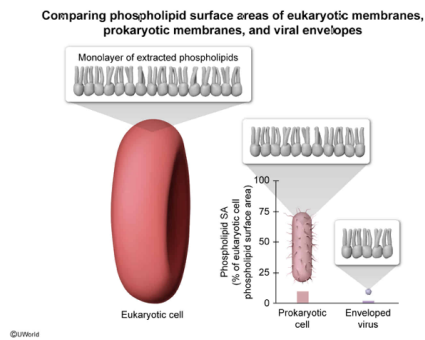


☒ D.



Correct
51% answered correctly

Explanation:



Eukaryotic cells, **prokaryotic cells**, and **viruses** exhibit many unique characteristics. For example, both eukaryotic and prokaryotic cells are surrounded by an outer plasma membrane whereas viruses may be either naked (ie, only surrounded by protein capsid) or enveloped (ie, protein capsid surrounded by an outer phospholipid bilayer). In addition, eukaryotic cells are the largest cells followed by prokaryotic cells, and viruses are generally **smaller** than either cell type.

In this question, phospholipid SAs were determined for multiple cell types and viruses using the methods described in the passage. The phospholipid SAs of a prokaryotic cell and enveloped virus were compared to the phospholipid SA of a eukaryotic cell.

Because prokaryotic cells are much *smaller* than eukaryotic cells, prokaryotic cells should also require *fewer* phospholipids to form their outer plasma membranes. Therefore, the SA of plasma membrane phospholipids in a prokaryotic cell should only be a *small* percentage of the SA of plasma membrane phospholipids in eukaryotic cells. In addition, the SA of phospholipids surrounding a smaller enveloped virus should be *less* than the phospholipid SA of either cell type. Only the graph in Choice D shows this result.

(Choices A and C) Because viruses are generally smaller (*not* larger) than prokaryotic cells, the SA of phospholipids making up the viral envelope should likely be smaller (*not* larger) than the SA of phospholipids making up the prokaryotic cell membrane.

(Choice B) Given their much *smaller* size, prokaryotes and viruses likely require a *much* lower amount of phospholipids to form their plasma membranes and envelopes, respectively. Therefore, the phospholipid SAs of the prokaryote and virus would likely be *much smaller* than the phospholipid SA of the eukaryotic cell. In addition, due to the very small size of most viruses, the phospholipid SA of the enveloped virus is likely *lower* than (*not* similar to) the phospholipid SA of the prokaryote.

Educational objective:

Eukaryotic and prokaryotic cells are surrounded by an outer plasma membrane. In contrast, viruses may be surrounded only by a protein capsid (ie, naked viruses) or have an outer phospholipid bilayer surrounding this protein coat (ie, enveloped viruses). In addition, although eukaryotic cells are larger than prokaryotic cells, viruses are generally smaller than both cell types.

Time spent: 0 sec
Subject:
Biology

QID: 403345
System:
Cellular Biology